

A LOW-POWER LOW-VOLTAGE BANDGAP REFERENCE IN CMOS

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ABSTRACT

A low-voltage low-power bandgap voltage reference in 90nm CMOS is designed and simulated. The overall bandgap architecture is optimized to achieve high accuracy temperature and power supply independent voltage reference. It consists of the bandgap core circuit, op-amp, start-up circuit and output stage. The bandgap reference circuit provides reference voltage of $584.7mV \pm 0.8mV$ with $1.2V \pm 10\%$ power supply and over $-40^\circ C$ to $125^\circ C$ temperature ranges simultaneously. The total layout area including dummy structures is $100\mu m \times 85\mu m$.

Index Terms— Bandgap Voltage Reference, analog IC, Low-Voltage, Low-Power, Op-Amp, Start-Up Circuit, Output Stage

1. INTRODUCTION

As the CMOS technology has developed rapidly during the past several decades and has brought us into a new era of high integration and ultra-low power consumption with the power supply voltage levels scaled down, the need for lower than the conventional bandgap voltage reference of $1.205V$ was created. Hence, it is critical to develop low-power consumption bandgap references working under low-voltage supply that is compatible with the other modern circuit blocks.

Since the first bandgap reference circuit based on Bipolar Junction Transistor (BJT) was introduced [1], the bandgap voltage reference circuit has been continuously improved in many aspects. A number of researchers are contributing new solutions based on the concept of using a fraction of the original bandgap voltage level [2, 3, 4] and for low-power supply voltage conditions [5, 6, 7].

In this paper, design and simulation of a high accuracy bandgap voltage reference circuit working over $0.9V$ to $1.4V$ power supply and $-40^\circ C$ to $125^\circ C$ temperature range in 90nm CMOS technology are presented. The rest of the paper is organized as follows. In Section 2 the basic concepts of building a bandgap reference are outlined. In Section 3, details of the proposed bandgap voltage reference circuit design are

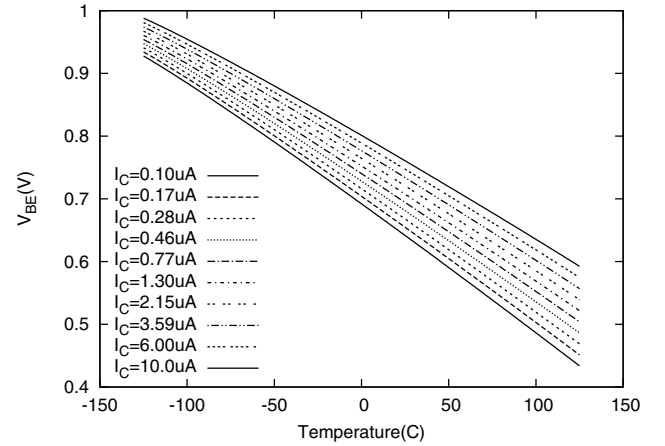


Fig. 1. Simulated base-emitter voltage V_{BE} dependence vs. the biasing current I_C and temperature.

presented. In Section 4, a simulated characterization results are provided for the bandgap circuit as well as the composing sub-blocks. The concluding remarks are given in Section 5.

2. TEMPERATURE INDEPENDENT REFERENCE

A typical BJT's base-emitter voltage V_{BE} has a negative temperature coefficient (TC), Fig. 1, and it is function of both temperature and the biasing current I_C . This particular property, the negative TC, of the V_{BE} voltage is not explicitly captured in the conventional approximated equation $V_{BE} = k/qT \ln(I_C/I_s)$, because temperature dependence of the saturation current I_s is much stronger and more complicated. In addition, it should be noted that the TC itself is also function of the collector current I_C .

It can be shown that base-emitter voltage V_{BE} dependence upon temperature T and the collector current I_C is somewhat more accurately captured by the following equation:

$$V_{BE}(I_C, T) = (a_1 \ln I_C + a_2)T + a_3 \quad (1)$$

where coefficients a_1, a_2, a_3 are technology dependent constants that can be determined by simulation. Obviously, con-

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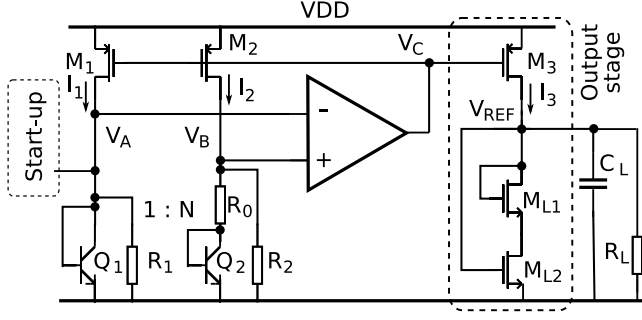


Fig. 2. Circuit diagram of bandgap voltage generator.

stant a_3 is the traditional $1.205V$ bandgap voltage, while expression $(a_1 \ln I_C + a_2)T$ is the temperature coefficient of V_{BE} that includes dependence upon the collector current I_C .

While a single base-emitter voltage V_{BE} exhibits negative TC, difference of two base-emitter voltages ΔV_{BE} , under condition that there are two different current densities used, exhibits positive TC. Hence, it is possible to deliver proportional to absolute temperature (PTAT) voltage that has positive TC [8]. Linear combination of these two voltages, therefore, may result in voltage generator with zero temperature coefficient, i.e.:

$$V_{REF} = \alpha_1 V_{BE} + \alpha_2 \Delta V_{BE} = \alpha_1 V_{BE} + \alpha_2 (V_T \ln n) \quad (2)$$

where α_1 and α_2 are design constants and n is the ratio between the two current densities.

3. CIRCUIT DESIGN

Temperature independent bandgap voltage lower than $1.205V$ can be realized by using a fraction of the original bandgap voltage, specifically, its current terms [3].

Schematic diagram of bandgap voltage generator, Fig. 2, shows the low-voltage bandgap core, the output stage and start-up node. In this configuration, the current through R_0 has positive TC, while the voltage value across R_1 and R_2 equals to the base-emitter voltage of the diode connected BJT transistor Q_1 . The current through M_2 (also M_1 , because $I_1 = I_2$) is temperature independent:

$$I_1 = I_2 = \frac{V_{BE}}{R_2} + \frac{V_T \ln N}{R_0} = \frac{1}{R_2} (V_{BE} + \frac{R_2}{R_0} V_T \ln N) \quad (3)$$

Value of current I_3 may vary slightly, following variations of the transistor M_3 size and the load impedance. The output reference voltage across the capacitive load C_L is

$$V_{REF} = I_3 R_L = k I_1 R_L \quad (4)$$

where $k = W_3/W_1$ is the $M_{3,1}$ transistor gate width ratio, and R_L is determined by the loading transistor $M_{L1,2}$ of the output stage.

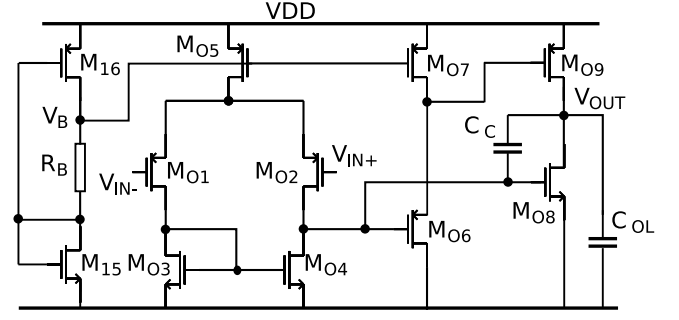


Fig. 3. Circuit diagram of two stage op-amp.

Table 1. Sensitivity of op-amp's voltage V_{OUT} .

Parameter	Case 1	Case 2
$(W/L)_{09}$	$1.6\mu m/0.8\mu m$	$2\mu m/1\mu m$
m	50	10
$V_{GS} - V_{th}$	$-10.7mV$	$-106.9mV$
I_D	$27.73\mu A$	$21.75\mu A$
$\Delta I_D (V_{DD} \pm 10\%)$	$\pm 20nA$	$-44nA$ $+24nA$
V_{out}	$812mV$	$517.5mV$
$\Delta V_{out} (V_{DD} \pm 10\%)$	$-17.74mV$ $+7.97mV$	$+0.3mV$ $+0.9mV$

Sensitivity to power supply variation: Two study cases presented in Table 1 demonstrate influence of gate-source voltage V_{GS} of M_{09} transistor on power supply variation sensitivity. The results imply that larger MOS transistor size is beneficial. In order to reduce circuit sensitivity to the nominal $\pm 10\%$ power supply disturbance, drain currents $I_{D1,2}$ should be kept as stable as possible. However, a three-way trade-off between power consumption, gate-source voltage V_{GS} and MOS size (W/L) must be very carefully balanced.

Op-Amp Circuit: Figure 3 shows circuit diagram of a classical two stage op-amp implementation that is used in this project. The first stage is a differential stage followed by the second stage composed of transistors M_{06} to M_{09} . Transistors M_{06} and M_{07} act as the level shifter while M_{08} and M_{09} act as a class AB push-pull output [9].

The V_{BE} voltage of $Q_{1,2}$ is set to approximately $695mV$ at room temperature. Hence, the input common-mode voltage of the op-amp is set to be in the $650mV \sim 750mV$ range. The op-amp is optimized to control the bandgap's feedback loop and it needs to have gain of more than $40dB$, Fig. 4.

The Biasing Circuit: Sub-circuit consisting of $M_{15,16}$ and R_B serves as a voltage divider, Fig. 3, which provides voltage reference that is dependent upon the power supply voltage. The biasing voltage V_B is $683mV$ for $1.2V$ power supply (nominal), and $600mV$ for $1.08V$ and $770mV$ for $1.32V$ power supply respectively.

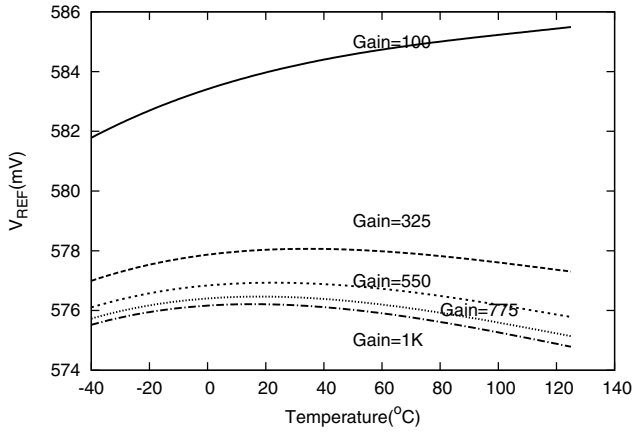


Fig. 4. Reference voltage V_{REF} vs. gain and temperature.

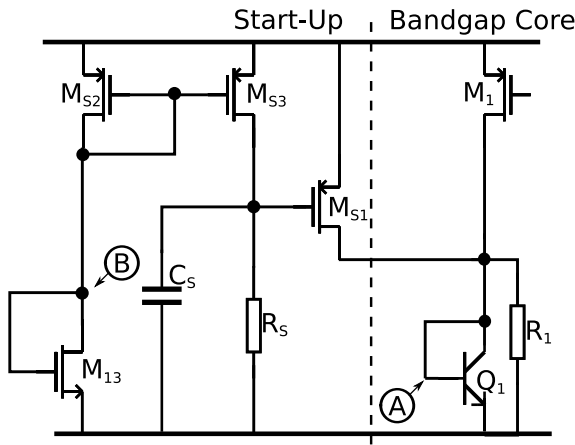


Fig. 5. Modified start-up circuit diagram.

Start-Up Circuit: Bandgap circuit has two stable operating points, in this case at $I_{C0} = 0$ and $I_{C0} = 8.8\mu A$, therefore a start-up circuit is needed to guarantee the desired mode of operation [3]. In a conventional start-up circuit, transistor M_{13} is BJT and its base is connected to the base of Q_1 (i.e. to node A instead of node B, Fig. 5) and capacitor C_S is not used. Under normal conditions and once the power supply is on, transistor M_{S1} injects a significant current into R_1 and Q_1 . At the end of the start-up phase, the current through M_{S3} makes the voltage across R_S high enough (close to V_{DD}) to turn off M_{S1} , thus the start-up circuit is turned off.

However, there are two issues with the conventional start-up circuit. First, a small percentage of the injected current flows into the base of M_{13} (when it is BJT with base connected to node A), which disturbs the balance in the bandgap core circuit due to the local feedback loop. Thus, the output reference voltage is also affected because there is a slight amount of current drawn from bandgap core circuit and then

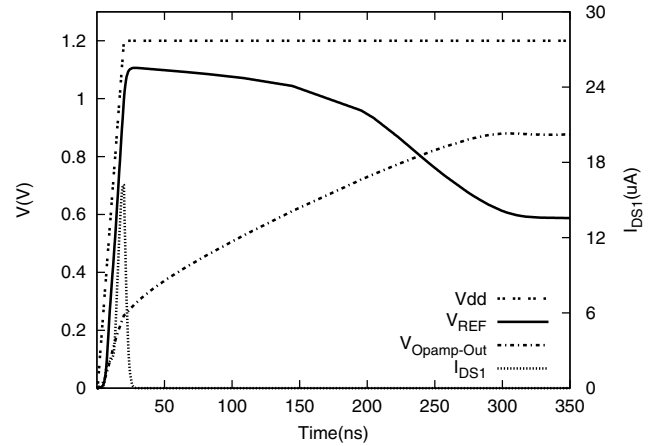


Fig. 6. The start-up transition period behaviour of V_{REF} , op-amp's output voltage, and M_{S1} drain-source current.

injected into the start-up circuit continuously, even when the circuit locks at the desired operating point. To solve this problem, M_{13} BJT is replaced by a diode connected NMOS transistor M_{13} , Fig. 5. This replacement guarantees that the start-up circuit is completely disconnected from the bandgap circuit once the start-up task is finished.

Second, drain current of M_{S1} may stop too early, even before the power voltage level is settled. To solve this problem, a gate capacitor C_S is added to delay the rise of the voltage at gate of M_{S1} . This delay also gives M_{S1} more time to turn on thoroughly and generate larger instantaneous current, Fig. 6.

The Output Stage: As shown in Fig. 2, M_3 mirrors the bandgap current through $M_{1,2}$ proportionally, while the current's absolute value is controlled by the main feedback loop. In order to reduce layout area and the temperature dependence of diffusion resistors, in the output stage appropriately biased transistors are used instead. Transistor M_{L2} is biased in linear region, while M_{L1} is diode connected and increases V_{REF} voltage by V_{thn} . The biasing voltage of M_{L2} is the output voltage itself, as this node generates the most stable voltage in the circuit (in respect to temperature and power supply voltage variations). This, the biasing arrangement drastically reduces dependance of the transistor's output impedance in respect to temperature and power supply voltage because the resistance of M_{L2} is controlled only by V_{GS} , which is equivalent to V_{REF} voltage. In addition, a $0.75pF$ C_L capacitor is connected as a capacitive load to the bandgap voltage reference output node.

3.1. Temperature dependance of passive components

The n-type and p-type diffusion resistors have the opposite sign temperature coefficients, which implies that by combining the two type of resistors in series with an appropriate ratio, the overall TC can be made very low. Considering that the

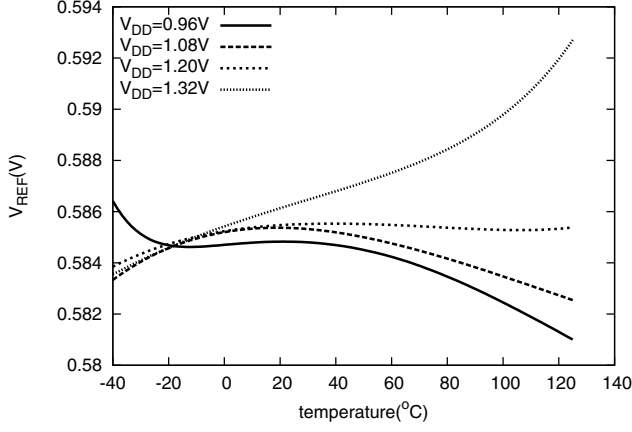


Fig. 7. Simulation graph for V_{REF} vs. V_{DD} and temperature.

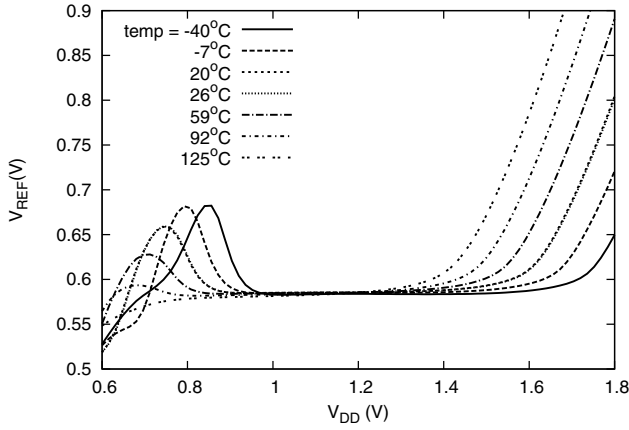


Fig. 8. Simulation graph for V_{REF} vs. temperature and V_{DD} .

bandgap reference voltage has a convex characteristic, use of real resistors have a slight complementary effect on the output voltage.

4. SIMULATION RESULTS AND DISCUSSION

Graph with family of temperature sweep curves for various power supply voltages, Fig. 7, shows that the reference voltage varies by $244\mu V$ over temperature sweep from $0^\circ C$ to $125^\circ C$ at the nominal V_{DD} voltage. Slight increases of V_{REF} at the low temperature and low power supply, and for high temperature and high power supply are caused by the gate-capacitors' temperature dependence.

To better investigate the power supply dependence, graph with family of curves of the output voltage dependence against the supply voltage sweep is shown in Fig. 8. The ripples in $600mV$ to $1V$ range are caused by the current difference be-

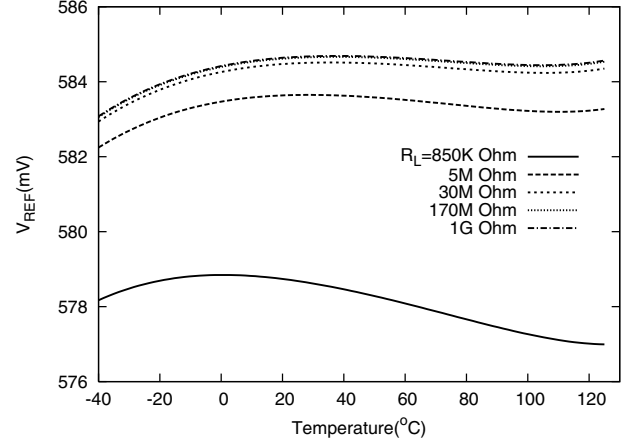


Fig. 9. Dependence of V_{REF} vs. load impedance and temperature.

tween the two bipolar transistors in the bandgap core circuit. When V_{DD} is low, it is easier for the bipolar transistors with lower current density to carry more current under the same condition of low V_{BE} . For high temperature, i.e. $125^\circ C$, the ripple disappears as the high temperature makes the bipolar transistor capable to work under higher current with the same low V_{BE} , which eliminates the current difference between the two bipolar transistors, as shown by this equation [10]

$$I_{BE}(T_2) = I_{BE}(T_1) \left(\frac{T_2}{T_1} \right)^{X_{T1}/n_F} \exp \left[-\frac{qE_g(300)}{n_F k T_2} \left(1 - \frac{T_2}{T_1} \right) \right]$$

In order to specify the lower limit of the resistive load, the bandgap output node is connected to a load resistor whose resistance is swept over wide range. The simulation results show that the resistive load has lower limit at around $850k\Omega$, under constrain that a variation of reference voltage V_{REF} is no more then $1.849mV$ across the temperature range of $-40^\circ C$ to $125^\circ C$, which means the error due to temperature variation is 0.3% , Fig. 9.

More then 1000 Monte Carlo Analysis runs were executed that covered all corner extremes. If all of these runs that generated $|error| \geq 1\%$ are counted as failure, then the success rate for process variation and mismatch is around 85% and the yield is expected to be over 90% . Table 2 shows summary of the simulation specifications. This bandgap achieved very high accuracy without additional compensation circuit and it is able to operate within a very wide temperature range. The main drawback is relatively high dependence on the supply voltage, which may be caused by the high demand of the the CMOS op-amp, i.e. the op-amp itself needs to work under very harsh low-voltage conditions.

Layout design occupies $100.92\mu m \times 84.74\mu m$ area, which includes dummy ring of BJT devices and array of decoupling capacitors, Fig. 10.

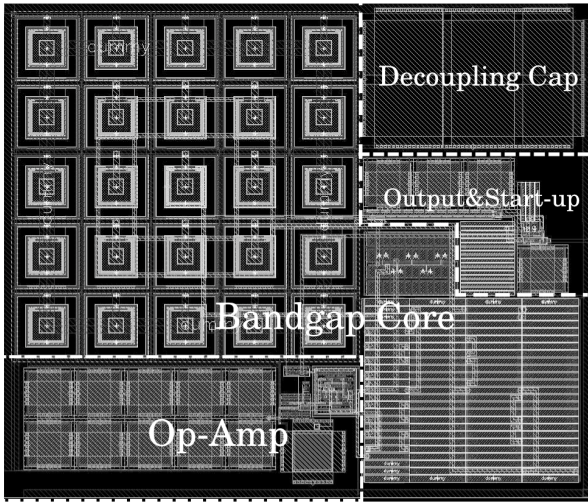


Fig. 10. The bandgap voltage reference layout.

5. CONCLUSIONS

A CMOS bandgap circuit has been presented for operation with 1.2V nominal power supply, while the extended power supply range is $0.96V \sim 1.4V$. The start-up block is further improved and an innovative output stage is applied, both contributing to improvement in performance and the accuracy. As a result, the presented circuit achieves very high accuracy of the voltage reference over wide temperature variation range and performs very well under wide power supply variation, without additional curvature compensation circuits. In the future work, contribution of curvature compensation circuit will be studied in more details.

6. ACKNOWLEDGEMENTS

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7. REFERENCES

- [1] R.J. Widlar, "New developments in IC voltage regulators," *Solid-State Circuits, IEEE Journal of*, vol. 6, no. 1, pp. 2–7, Feb 1971.
- [2] H. Banba, H. Shiga, A. Umezawa, T. Miyaba, T. Tanazawa, S. Atsumi, and K. Sakui, "A CMOS Bandgap Reference Circuit with Sub-1-V Operation," *Solid-State Circuits, IEEE Journal of*, vol. 34, no. 5, pp. 670–674, May 1999.

Table 2. Summary of simulation specifications.

Power Supply Voltage	$1.2V \pm 10\%$
Technology	90nm CMOS
Bandgap Cell Area	$100\mu m \times 85$
Power Consumption @ $T = 25^\circ C$	$160\mu W$
Temperature Range	$-40^\circ C \sim 125^\circ C$
V_{REF} @ $T = 25^\circ C$	$584.7mV$
V_{REF} Temperature Variations (Without Curvature Compensation)	$244.37\mu V, \quad 3.343ppm/^\circ C$ ($0^\circ C \sim 125^\circ C$) $1.6119mV, \quad 16.7ppm/^\circ C$ ($-40^\circ C \sim 125^\circ C$)
V_{REF} Variations vs. Power Supply Voltage	$1.1mV$ ($1.08 \sim 1.32V_{DD}$)
Thermal noise	$1.4604 \times 10^{-16} / Hz$
PSRR	20.685 dB

- [3] P. Malcovati, F. Maloberti, C. Fiocchi, and M. Pruzzi, "Curvature-compensated BiCMOS Bandgap with 1-V Supply Voltage," *Solid-State Circuits, IEEE Journal of*, vol. 36, no. 7, pp. 1076–1081, Jul 2001.
- [4] A. Cabrini, G. De Sandre, L. Gobbi, P. Malcovati, M. Pasotti, M. Poles, F. Rigoni, and G. Torelli, "A 1 V, 26 uW Extended Temperature Range Band-gap Reference in 130-nm CMOS Technology," in *Solid-State Circuits Conference, 2005. ESSCIRC 2005. Proceedings of the 31st European*, Sept. 2005, pp. 503–506.
- [5] Johan H. Huijsing, Rudy J. van de Plassche, and Willy M. C. Sansen, *Analog Circuit Design, Low-Noise, Low-Power, Low-Voltage; Mixed-Mode Design with CAD Tools; Voltage, Current and Time References*, Kluwer Academic Publishers, 1996.
- [6] Willy M. C. Sansen, *Analog Design Essentials*, Springer, second edition, 2006.
- [7] Ka Nang Leung and P. K. T. Mok, "A sub-1-V 15-ppm/C CMOS bandgap voltage reference without requiring low threshold voltage device," *Solid-State Circuits, IEEE Journal of*, vol. 37, no. 4, pp. 526–530, Apr 2002.
- [8] Behzad Razavi, *Design of Analog CMOS Integrated Circuits*, Mc Graw Hill Education, 2001.
- [9] Roubik Gregorian, *Introduction to CMOS Op-Amps and Comparators*, John Wiley and Sons, Inc., 1999.
- [10] Giuseppe Massabrio and Paolo Antognetti, *Semiconductor Device Modeling with Spice*, McGraw-Hill Professional, 01 edition, 1998.